

CBCT

Three dimensional dental imaging

Cone beam computed tomography shows your teeth and jawbone in three dimensions. It helps us understand important anatomical relationships and plan treatment when an examination and two dimensional X-rays do not provide enough information.

What the third dimension adds

Multiple X-ray projections are reconstructed into a digital 3D data set. We can view thin slices in several directions and assess distances and spatial relationships. In German, this examination is called **dentale Volumentomographie (DVT)**.

Detailed images, a short scan and focused planning. Teeth and bone can be assessed without the overlap seen on a two dimensional image. The scan is painless and the digital images are quickly available for interpretation.

Common uses in our practice

Wisdom teeth and the nerve canal

If the relationship of wisdom tooth roots to the nerve canal is unclear, especially in the lower jaw, CBCT can support surgical planning. It mainly shows the bony canal, rather than the nerve itself.

Impacted and displaced teeth

Assess the position relative to nearby teeth, possible root damage and the access for surgical exposure or attachment of a traction chain. This applies to different teeth, not only canines.

Implants and bone grafting

Assess bone height and width, the nerve canal and the maxillary sinus. Where planned, the data can be combined with a surface scan to produce a surgical guide.

Root canal treatment and apical surgery

For unexplained symptoms or complex cases: investigate additional canals, inflammation around root tips, root resorption or problems after root canal treatment, and plan root-end surgery.

The clinical question comes first: CBCT is not automatic routine imaging before every implant, wisdom tooth procedure or orthodontic treatment. The additional information must be relevant to your care.

Further uses of 3D imaging

In each case, the additional spatial information must be expected to benefit diagnosis or treatment.

Inflammation and bone changes

Assess the extent of a cyst, an unexplained bone lesion, bone infection or osteonecrosis of the jaw. Suspected tumours may require other imaging and a tissue sample.

Maxillary sinus and foreign bodies

Investigate a possible link between teeth and sinus symptoms, locate foreign bodies or assess difficult anatomy before a sinus lift.

Dental injuries and bone defects

Assess tooth roots and surrounding bone after an accident if the findings remain unclear. Selected complex defects around teeth may also benefit from spatial assessment.

Orthodontics and the jaw joint

Plan selected complex tooth movements, orthodontic mini-screws or jaw corrections. CBCT can show bony joint changes; MRI is usually more suitable for the joint disc.

Checking for dental infection before antiresorptive treatment

Before treatment with **bisphosphonates** or **denosumab**, it is important to identify dental and jaw infections that need treatment. These medicines are used, for example, for osteoporosis or cancer and can increase the risk of osteonecrosis of the jaw.

The starting point is a dental examination with appropriate X-rays. Existing images, small dental X-rays or a panoramic image may be sufficient. CBCT can add information if a specific question remains unanswered, such as the extent of an infection.

■ **CBCT is not a routine screening requirement for everyone.** A normal scan cannot rule out future jaw
■ osteonecrosis. Tell us the medicine, reason for treatment and planned or actual start date. Where
■ needed, we coordinate dental care and the start of medication with your treating doctors. Do not stop
■ medication on your own.

What CBCT cannot do

Metal, root fillings and movement can cause image artefacts. Very fine cracks or early changes may remain undetected. CBCT does not automatically explain every cause of pain and cannot guarantee complication-free treatment. Soft tissues, extensive infections or certain tumour questions may require CT, MRI or further investigations.

CBCT compared with medical CT

The clinical question and the structure being examined determine the appropriate technique.

Aspect	Dental CBCT	Medical CT
Teeth and bone	Can offer very high spatial resolution, especially with small imaging fields.	Also suitable for bone assessment; image quality depends on the protocol.
Radiation dose	Often lower than medical CT for a comparable dental question.	Depends on equipment and protocol; modern low-dose CT and CBCT dose ranges can overlap.
Soft tissues	Limited assessment; usually without contrast medium.	Better soft tissue assessment, with contrast medium when needed, for example for extensive infections.
Availability	Can be integrated into care at our practice; digital images are quickly available.	Usually arranged through a radiology provider; CT scans themselves are also short.

CT as an alternative – an individual decision

CT cannot replace CBCT for every indication. A routine CT protocol may have insufficient spatial resolution for fine dental detail; CT has advantages for soft tissues. Metal and movement can affect either method. The extent of artefacts depends on the equipment, protocol and region examined.

If CT is suitable for your clinical question, we will explain this individually and coordinate the examination with radiology where needed. Existing images are reviewed first. CT information from ACR/RSNA.

Radiation protection and alternatives

CBCT uses **ionising radiation**. A small additional long-term radiation risk cannot be excluded. The expected benefit must outweigh this risk. A small dental X-ray or panoramic image usually involves a lower dose and may already answer the question.

We review existing images and choose the smallest adequate imaging field and an appropriate programme. We take particular care with children and adolescents. **Tell us before the scan if you are or could be pregnant.** We then consider alternatives, postponement or, if urgently needed, an individually justified examination.

Costs of €180 to €280

CBCT costs €180–280 depending on the indication and the necessary scope of the examination.
We explain the specific fee and included services before the scan.

In German dental care, CBCT is generally charged as an **individually paid private service** for patients with statutory insurance. Reimbursement by private insurance, civil service benefits or supplementary insurance depends on the policy and medical justification. A referral does not guarantee payment.

What happens at your appointment

1. **Prepare:** Bring your referral, previous images and medication list, or arrange for them to be sent in advance. Bring a planning template only if we have specifically requested it.
2. **Position:** We carefully position your head. Remove glasses, jewellery and removable dental appliances as instructed. Fixed braces and implants remain in place.
3. **Scan:** The machine moves around your head. The scan is brief. Please keep still and follow our instructions. Tell us beforehand if positioning or keeping still is difficult.
4. **Interpret:** We assess the entire recorded volume. Incidental findings may need further investigation. We discuss the results and arrange access to the report and image data for your continuing care.

For CBCT alone, anaesthesia, injections and contrast medium are usually unnecessary. You do not need to fast. Separate instructions apply if you are also having surgery or sedation.

Frequently asked questions

When can I return to everyday activities

Immediately after CBCT alone: eating, drinking and driving are normally possible. No radiation remains in your body. You do not usually need someone to collect you for the scan alone.

Can existing scans be used

Yes. Please provide the original digital image data and report where possible. We check whether their age, quality and imaging field are suitable. A repeat scan requires a new justification.

What if I do not want a CBCT scan

We discuss alternatives and any remaining uncertainty. The planned treatment may need to be modified or postponed.

Your appointment checklist

- The clinical question, benefit and costs have been discussed.
- Your referral, previous images and medication list are available.
- Possible pregnancy and difficulty keeping still have been discussed.
- Result discussion and continuing treatment have been arranged.

Questions: 06151 786540 · info@mkg-im-carree.de. [Current online CBCT information](#).

For problems after an additional procedure outside practice hours: +49 151 10437450; if unavailable, dental emergency service 01805 60 70 11. Call 112 for a life-threatening emergency, breathing difficulty, severe difficulty swallowing or rapidly increasing swelling of the floor of the mouth, neck or tongue.

Clinical sources: [AWMF 083-005, S2k Dentale digitale Volumentomographie \(12/2022, valid until 12/2027\)](#); [FDA, Dental CBCT](#); [ACR/RSNA, Dental Cone Beam CT](#); [AAOMS, MRONJ Position Paper \(2022\)](#); [§ 120 StrlSchV](#). Accessed 24 September 2026.

This information supports your individual consultation.